Summary in Graph

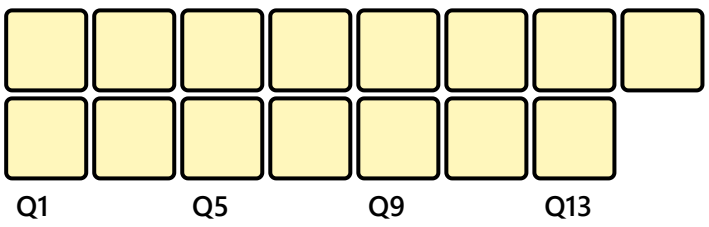
Exam Summary (GO Classes CS | Digital Logic | Boolean Algebra | Weekly Quiz 3)

Qs. Attempted:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)
Correct Marks:	0 (0 + 0)
Penalty Marks:	0 (0 + 0)
Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (31) out of 314 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Digital Logic

For 4 input boolean variables, the number of boolean functions possible is ?

Your Answer: | Correct Answer: 65536 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

For the identity $AB + A'C + BC = AB + A'C$, the dual form is

A. $(A+B)(A'+C)(B+C) = (A+B)(A'+C)$

B. $(A'+B')(A'+C')(B'+C') = (A'+B')(A+C')$

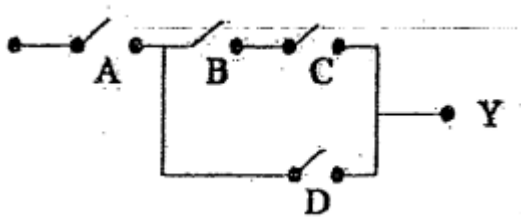
C. $(A+B)(A'+C)(B+C) = (A'+B')(A+C')$

D. $A'B' + AC' + B'C' = A'B' + AC'$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

For the switch circuit, taking open as 0 and closed as 1, the expression for the circuit is Y.



A. $A + (B+C)D$

B. $A + BC + D$

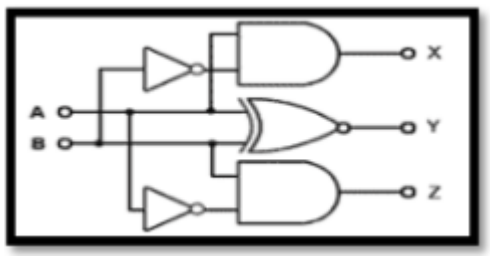
C. $A(BC + D)$

D. None of these

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

Refer to the following figure. If $\mathbf{A} = \mathbf{0}$ and $\mathbf{B} = \mathbf{1}$, what will be the logic states at $\mathbf{X}, \mathbf{Y}$ and $\mathbf{Z}$?



A. $\mathrm{X} = 1, \mathrm{Y} = 1, \mathrm{Z} = 0$

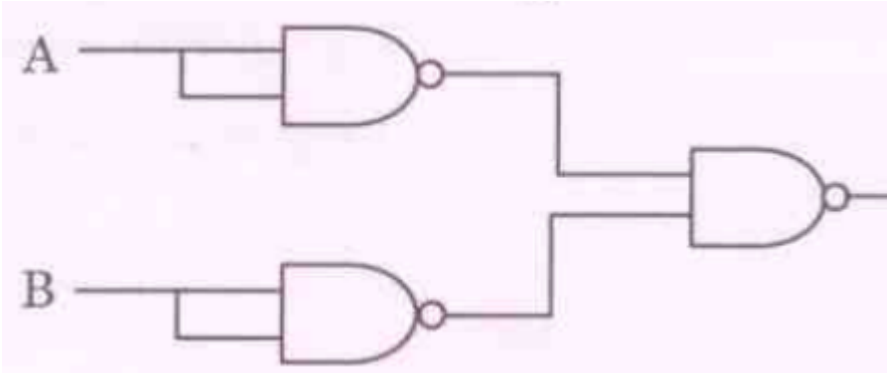
B. $\mathrm{X} = 1, \mathrm{Y} = 0, \mathrm{Z} = 0$

C. $\mathrm{X} = 0, \mathrm{Y} = 0, \mathrm{Z} = 1$

D. $\mathrm{X} = 0, \mathrm{Y} = 1, \mathrm{Z} = 0$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic



Given circuit represents

A. INVERTER

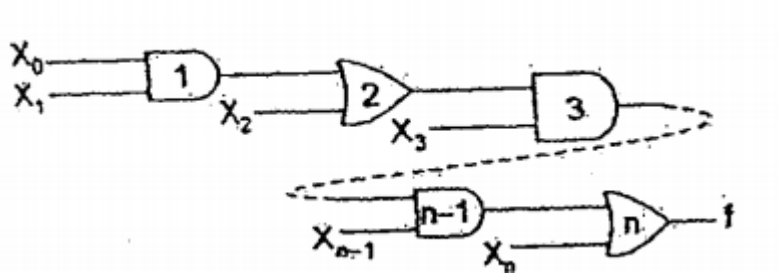
B. AND

C. OR

D. NOR

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic



In the given network of AND and OR gates **f** can be written as:

A. $X0X1X2.....Xn + X1X2.....Xn + X2X3....Xn....Xn$

B. $X0X1 + X2 + X3 + ...+Xn - 1.Xn$

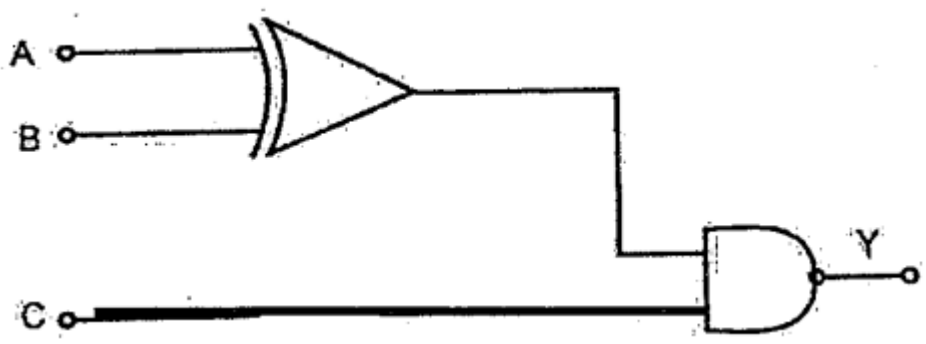
C. $X0 + X1 + X2 + ...+Xn$

D. $X0X1X3.....Xn - 1 + X2X3X5.....Xn - 1 + ...+Xn - 2Xn - 1 + Xn$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The Boolean expression for the output of the logic circuit shown in the figure is



A. $Y = AB + A'B' + C$

B. $Y = AB + A'B' + C'$

C. $Y = AB' + A'B + C$

D. $Y = AB + A'B + C'$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Boolean algebra has an interesting property called duality.
The dual form of $\mathrm{A} + \mathrm{BC}$?

A. $\mathrm{AB} + \mathrm{C}$
B. $A'B' + C'$
C. $A(B + C)$
D. $A'(B' + C')$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

Some of the theorems of Boolean algebra are not true for ordinary algebra. Similarly, some of the theorems of ordinary algebra are not true for Boolean algebra.
Which of the following cancellation laws is/are not true for Boolean algebra ?

A. If $\mathrm{A} + \mathrm{B} = \mathrm{A} + \mathrm{C}$, then $\mathrm{B} = \mathrm{C}$
B. If $\mathrm{AB} = \mathrm{AC}$, then $\mathrm{B} = \mathrm{C}$
C. If $\mathrm{A} \oplus \mathrm{B} = \mathrm{A} \oplus \mathrm{C}$, then $\mathrm{B} = \mathrm{C}$
D. If $\mathrm{A} \odot \mathrm{B} = \mathrm{A} \odot \mathrm{C}$, then $\mathrm{B} = \mathrm{C}$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 02sec | Discuss

Q #10 Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

Some of the theorems of Boolean algebra are not true for ordinary algebra. Similarly, some of the theorems of ordinary algebra are not true for Boolean algebra.
Which of the following is/are true for Boolean algebra?

A. If $x + y = x + z$, then $y = z$
B. If $xy = xz$, then $y = z$
C. If $\mathrm{y} = \mathrm{z}$, then $\mathrm{x} + \mathrm{y} = \mathrm{x} + \mathrm{z}$
D. If $y = z$, then $xy = xz$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

Which of the following statements are always true, in Boolean algebra?

A. If $A + B = C$, then $AD' + BD' = CD'$

B. If $A'B + A'C = A'D$, then $B + C = D$
C. If $A + B = C$, then $A + B + D = C + D$
D. If $A + B + C = C + D$, then $A + B = D$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

In a Boolean algebra, which of the following statements are true ?

A. If $x\,(y + a') = x\,(y + b')$, then $a = b$.
B. If $a'b + ab' = a'c + ac'$, then $b = c$.
C. $A \oplus BC = (A \oplus B)(A \oplus C)$
D. $x + y = x \oplus y \oplus xy$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

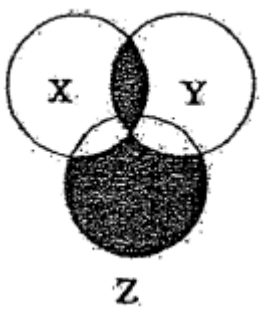
Q #13 Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

In a Boolean algebra, which of the following statements are true?

A. $x(y \oplus z) = xy \oplus xz$
B. $x + (y \oplus z) = (x + y) \oplus (x + z)$
C. $x(y \equiv z) = xy \equiv xz$
D. $x + (y \equiv z) = (x + y) \equiv (x + z)$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic



The Boolean expression for the shaded area in the Venn diagram is

A. $X'+Y'+Z$

B. $XY'Z + X'YZ$

C. $X + Y + Z$

D. $X'Y'Z+XY$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The simplification of the Boolean expression $(\overline{\overline{A}B\overline{C}}) + (\overline{A\overline{B}C})$ is

A. 0

B. 1

C. A

D. BC

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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